Gopika Jaideep

AI / ML Engineer | LLM Applications, RAG, Agentic Systems

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Summary

AI/ML engineer with a Master of Artificial Intelligence, specialising in end-to-end LLM applications, multi-agent systems and hybrid RAG, with Python and FastAPI backends on serverless AWS and React and Vue front ends, delivered in agile teams and to production. Seeking a full-time AI/ML engineering role in Australia.

Career history

AI Specialist at Timeless Tributes AI, Melbourne
Apr 2026 to Sept 2026

- Built PhotoAlchemy, a production photo-restoration and animation tool (React, Vite, Replicate API, GFPGAN, Real-ESRGAN, Nano Banana) with low-latency image processing
- Built LiveFlow, a funeral-director-facing PWA for livestreamed ceremony playback
- Built an AI video-generation prompt and credit-efficiency system spanning four platforms (Runway, OpenArt, Kling and ElevenLabs)
- Drafted outreach to around 150 independent funeral directors and ran QA, SOPs and production trackers

Data Project Lead (Volunteer) at AusWise, Melbourne
Mar 2026 to Present

Promoted from Data Project Coordinator to Data Project Lead.

- Built a social media analytics pipeline (Instagram scraping via Apify, XGBoost and SHAP, BERTopic, Groq LLM summarisation) producing branded performance reports
- Lead a 7-member data team, working closely across HR, employability, public health, marketing and project management, to deliver automated LLM-driven reporting
- Organised the AusWise Walkathon 2026 (370+ registrants), planned VIP invitations for the Safe and Sound fundraiser at Library at The Dock, and manage award-submission documentation

Retail Volunteer at Australian Red Cross, Collingwood
Mar 2025 to Present

Shop operations and customer service.

Data Science Intern at Verzeo
Nov 2020 to Dec 2020

Data cleaning and visualisation in Python.

Cybersecurity Intern at Verzeo
Sept 2020 to Oct 2020

Hands-on work with Kali Linux, Wireshark and Nmap.

Key projects

CareWise | Live app | Repository
Self-initiated project

AI companion for cancer caregivers, tracking symptoms, medications and appointments alongside emotional support.

- Built a six-agent orchestration layer (FastAPI, LLM-based intent routing, agent handoffs with cycle detection), implemented directly against provider SDKs (Gemini, Anthropic) without an agent framework
- Built a hybrid RAG pipeline (BM25 plus embeddings, reciprocal rank fusion) for a cited resource guide, restricting displayed citations to sources actually retrieved
- Designed layered crisis detection: deterministic keyword rules plus a parallel AI risk screen that can only add protection, with distinct handling when a carer reports risk to someone else

- Set up an eval-gated CI pipeline that blocks merges on crisis-detection or retrieval regressions, backed by 39+ Pytest tests
- Found and fixed two production bugs by testing the system in degraded states (AI layer down, compound multi-intent messages)

TraceAI | Repository | Demo

Self-initiated project

Missing-persons lead-scoring and evidence-fusion system for law enforcement and NGO caseworkers.

- Built a four-signal fusion engine (location, recency, credibility, identity) that renormalises weights for missing signals and applies a physical-plausibility cutoff
- Wrote a mutation-testing harness that deliberately breaks four safety controls (access gating, attestation, honeypot, audit chain) and checks the 48-test suite catches each

Quitory | Frontend repository

Monash Master of AI industry studio project, team of 3, selected for the Monash Industry Showcase

Vaping-cessation support app on a serverless AWS backend.

- Owned six backend modules on serverless AWS (Lambda, API Gateway, RDS MySQL)
- Built a Groq-backed conversational assistant with keyword-gated topic scoping, and a hybrid rule-based and LLM health-risk calculator using the WHO Global Health Observatory API
- In the evaluation phase, selected the best of 25 prompt variants on the 75% training portion of 2,500 examples, then improved embedding-similarity relevance by 38 percentage points over a GPT-4 baseline on the held-out 25% test split
- Red-teamed with 500+ adversarial queries, with 0% unsafe outputs on the tested queries

Earlier projects

- **ParkingSpy (capstone):** data pipelines over 6 live data sources feeding a Random Forest model reaching 94.2% accuracy, cutting detection latency from about 8 minutes to under 30 seconds, delivered in an agile, sprint-based team
- **Network intrusion detection (B.Tech capstone):** compared Random Forest, SVM and LSTM on NSL-KDD (Random Forest 99.9% accuracy on a random 70/30 split) and built an LSTM web demo tested with simulated MQTT attacks
- **ALOKA (B.Tech project):** OpenCV-based Android accessibility app built in Java and Kotlin (repository)

Education

Master of Artificial Intelligence from Monash University, Melbourne
Feb 2024 to Dec 2025

Bachelor of Technology, Computer Science and Engineering from SCMS School of Engineering and Technology (SSET), Kochi, India, affiliated to APJ Abdul Kalam Technological University (KTU)
2019 to 2023

Certifications

Professional Certification in Data Engineering, GlofAI; Machine Learning with Python, Anaconda; Building AI-Powered Chatbots, IBM; Advanced Python, LinkedIn Learning

Key skills

AI and ML: Hybrid RAG (BM25 and embeddings, reciprocal rank fusion), multi-agent orchestration, LLM evaluation, eval-gated CI, mutation testing, PyTorch, distributed training fundamentals, scikit-learn, XGBoost, SHAP, BERTopic, OpenCV
Languages and frameworks: Python, FastAPI, TypeScript, React, Vue, Kotlin, Java (Android), C and C++ (fundamentals)
Cloud and data: AWS (Lambda, API Gateway, RDS, EC2, S3), GCP (hands-on deployment), Databricks, Apache Spark and PostgreSQL (Data Engineering certification), SQL, MySQL
APIs and tools: Anthropic, Gemini, Groq and Replicate APIs, WHO Global Health Observatory API, Apify, gTTS, async I/O and multiprocessing, Git, Pytest, CI/CD, Kali Linux, Wireshark, Nmap
Spoken languages: English, Hindi, Malayalam (fluent), Tamil (functional)